THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student sprinkles water on a hot surface and it disappears. The process is: 1

(a) condensation **(b)** freezing **(c)** evaporation **(d)** melting

2. Water in a closed pot of a pressure cooker boils above 100 degrees Celsius because: 1

(a) increased pressure raises the boiling point
(b) pressure is lower
(c) heat is lost
(d) water is heavier

3. A gas in a syringe is compressed. The pressure of the gas: 1

(a) becomes zero **(b)** decreases **(c)** increases **(d)** stays same

4. Naphthalene balls kept in a cupboard get smaller over time due to: 1

(a) condensation **(b)** sublimation **(c)** melting **(d)** freezing

5. On high mountains water boils below 100 degrees Celsius since: 1

(a) atmospheric pressure is lower **(b)** air is warm
(c) water is salty **(d)** the temperature is higher

6. Boiling water in an open pan keeps boiling at about 100 degrees Celsius even when more heat is supplied because: 1

(a) heat is lost
(b) the extra heat is used to change water into steam
(c) bubbles stop
(d) water becomes cold

7. A steel spoon and a plastic spoon are both in the: 1

(a) plasma state **(b)** solid state **(c)** gaseous state **(d)** liquid state

8. Which has the highest kinetic energy of particles at the same mass? 1

(a) water at 20 degrees Celsius **(b)** water at 50 degrees Celsius
(c) steam at 100 degrees Celsius **(d)** ice at 0 degrees Celsius

9. A liquid X evaporates faster than liquid Y at the same temperature. We can say: 1

(a) forces in X are stronger **(b)** Y is a gas
(c) forces between particles of X are weaker **(d)** X is a solid

10. The cooling effect of evaporation is maximum in a: 1

(a) closed room **(b)** humid place **(c)** cold place **(d)** dry, windy place

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a sealed packet of chips swell on a hill? 2

12. Why does a cold glass of water have droplets outside on a humid day? 2

13. Why does the smell of hot food travel faster? 2

14. Why do people use steam to cook idlis rather than hot water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how an earthen pot keeps water cool in summer. 3

16. Two liquids A and B boil at 78 degrees Celsius and 100 degrees Celsius. Which has stronger forces between particles? Explain. 3

17. Explain why the temperature stays constant when water freezes at 0 degrees Celsius although heat is being removed. 3

OR

Compare steam and water, both at 100 degrees Celsius, in terms of energy and particles. 3

18. Explain how camphor placed in a closed bottle gradually disappears. 3

OR

Why do deodorant sprays make the can cold? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain what happens to the heat supplied during melting and boiling, with reference to steam burns and the cooling of a drink by ice. 5

OR

Describe the role of the particle model in explaining diffusion, pressure of gases, evaporation and sublimation. 5

20. Explain why a mixture of ice and water at 0 degrees Celsius cools a drink more effectively than water at 0 degrees Celsius. Discuss in terms of heat flow. 5

OR

Explain how humidity, temperature, wind speed and surface area affect the rate of drying of clothes, and design a fair test for one of them. 5

END OF QUESTION PAPER — Total: 40 Marks

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